

PARAMJEET ROUT

Gurugram, India (Open to Remote / Pan-India) | +91 9337718142 | routparamjeet@gmail.com
[linkedin.com/in/paramjeet-rout2026](https://www.linkedin.com/in/paramjeet-rout2026) | github.com/routparam12 | substack.com/@paramjeetrout

Professional Summary

Generative AI Engineer and Data Scientist with 2.8+ years delivering **production-grade Generative AI, Large Language Model (LLM)** integrations, and **Retrieval-Augmented Generation (RAG)** pipelines using LangChain, LangGraph, LlamaIndex, Hugging Face, FAISS, and ChromaDB. Built two production AI systems: **AI Agent–Powered Customer Support Assistant** (multi-agent orchestration, Redis, Azure CI/CD) and **PaperShield** (leak-proof AI exam engine). Applied ML for fraud detection with **70% precision**. Strong foundation in **OLTP/OLAP, ETL pipelines, SQL, Root cause analysis** and Power BI.

Experience

Data Scientist (GenAI) — Unimrkt Response LLP

Oct 2025 – Present | Gurugram (Hybrid)

- **Architecting production RAG chatbot** — replaced static replies with a **multi-agent AI pipeline** (LangChain + LangGraph) retrieving context via FAISS vector search before LLM generation, enabling context-aware Natural Language Processing (NLP) at production scale.
- **Designed OLTP-to-OLAP data architecture** — PostgreSQL ingested into Elasticsearch via Logstash ETL with 3-minute incremental refresh, enabling 3-5 years of data warehouse analysis without impacting the source DB.
- **Delivered 13 end-to-end Power BI and Elasticsearch dashboards** backed by 50+ validated SQL pipelines, reducing reporting time by **94%** through automated ETL and optimised data models.
- **Built a fraud detection system** using fuzzy matching and Decision Tree classification (scikit-learn) — achieving **70% precision**; tracked via MLflow for model evaluation (precision, recall, and F1-score).
- Applied **token optimization** and chunk sizing strategies to balance LLM accuracy and API cost in a live production environment.
- DevOps pipeline: Dockerized service, Azure CI/CD (azure-pipelines.yml), JWT authentication—production-ready beyond prototype stage

Data Scientist Associate — Tech Mahindra

Mar 2024 – May 2025 | Ahmedabad

- Developed workflow automation scripts for PDF parsing, metadata extraction, and data cleaning pipelines using **Python + Azure Functions**
- Implemented role-based access retrieval for sensitive documents using **JWT authentication + FastAPI middleware**. **Integrated OpenAI, Gemini, and Together AI** into a unified abstraction layer enabling model fallback and cost-optimized inference
- Designed internal dashboards for usage analytics, retrieval accuracy tracking, and latency monitoring using **Streamlit + MLflow**. Collaborated directly with business teams to convert manual workflows into AI-assisted pipelines for measurable productivity gains.
- **Built Power BI dashboards** tracking Amazon product team KPIs, enabling client leadership to monitor delivery and quality metrics through self-serve analytics.
- **Reduced manual reporting effort by 30%** through optimised DAX measures and data models; wrote SQL extraction queries supporting multiple concurrent project reporting pipelines.

Projects

AI Agent– Powered Customer Support Assistant | *FastAPI · LangChain · LangGraph · Hugging Face · FAISS · Redis · PostgreSQL · Docker · Azure Pipelines*
Oct, 2025 – Dec, 2025

github.com/routparam12/SurveySenseAI

- **Built production Generative AI platform** with **multi-agent AI orchestration** via LangGraph, Hugging Face model integration, and FAISS vector store for fast similarity-based retrieval.
- **Designed a unified LLM abstraction layer** supporting OpenAI GPT and Google Gemini APIs with **model fallback** — enabling cost-optimized inference and zero-downtime model switching.
- **Implemented 9-type persistent memory** (SHORT_TERM, LONG_TERM, ENTITY, FACT, PREFERENCE, PROCEDURAL, EPISODIC, SEMANTIC, WORKING) stored in PostgreSQL via SQLAlchemy ORM with Alembic migrations.

- Integrated **Redis caching** for low-latency memory retrieval; multi-source ingestion (PDF, DOCX, email, text) feeding FAISS vector store for RAG; full DevOps: Docker, Azure CI/CD, JWT authentication.

PaperShield — AI Exam Engine for Leak-Proof Assessments | *FastAPI · LangChain · LangGraph · LlamaIndex · ChromaDB · Google Gemini API* *Apr, 2026 – Present*

github.com/routparam12/PaperShield-AI-Exam-Engine-for-Leak-Proof-Assessments

- **Eliminated paper leaks structurally** — MCQs generated at exam time via LlamaIndex + ChromaDB RAG pipeline; questions never stored in any database, making pre-exam leaks impossible.
- Applied **token optimization, model evaluation**, and LangGraph agentic AI orchestration; paginated 20-question delivery with shuffle logic; designed for JEE/NEET/State PSC government scale.

Fraud User Detection System | *Python · scikit-learn · Fuzzy Logic · Decision Tree · MLflow · Streamlit · Azure* *Feb, 2026 – Apr, 2026*

Azure deployment — company data, not publicly available

- **Built ML fraud detection pipeline** combining fuzzy matching (phonetic intelligence + edit-distance scoring) with Decision Tree classification to flag duplicate and high-risk accounts from call data.
- **Achieved 70% precision** (7/10 flagged accounts confirmed fraud); feature engineering on email similarity, IP geolocation, and behavioral signals; MLflow tracking and Streamlit dashboard for model monitoring.

Technical Skills

Generative AI & LLM: LangChain, LangGraph, LlamaIndex, Hugging Face, RAG Pipelines, Agentic AI, Multi-Agent Systems, Prompt Engineering, OpenAI GPT, Google Gemini API, Token Optimization, Model Fallback

Vector & Search: FAISS, ChromaDB, Elasticsearch, Pinecone, Vector Database, Semantic Search, Embedding Services

ML & NLP: scikit-learn, Decision Tree, Fuzzy Matching, Feature Engineering, Model Evaluation (Precision/Recall/F1), MLflow, Natural Language Processing (NLP)

Backend & Data: FastAPI, Python, PostgreSQL, SQLAlchemy, Redis, Docker, Azure Pipelines, Git/GitHub, Streamlit

Data Engineering: OLTP, OLAP, ETL Pipelines, SQL Optimization, Logstash, Data Warehouse, Power BI, DAX, Pandas

Certifications

- **Anthropic Academy** — Building with the Claude API (2026)
- **Microsoft PL-300** Power BI Data Analyst — *In Progress (2026)*
- **Microsoft DP-900** Azure Data Fundamentals — *In Progress (2026)*
- Google Antigravity Program — Generative AI & Applied AI Concepts (2025)
- RAG Learning & Bot Building — Hands-on Completion (2025)

Leadership and Open-Source Contributions – OSCG'26

- Supervised contributors reviewed and merged 30+ PRs across AI and automation-based projects.
- Mentored multiple teams in implementing clean, maintainable code and robust AI-driven systems.

Education

B.Tech — Computer Science & Engineering — **Gandhi Institute of Excellent Technocrats (GIET), Bhubaneswar**
2020 – 2024 | CGPA: 8.4 / 10